Superstore Marketing Campaign Analysis

Predicting Gold Membership Acceptance

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Career 247 — Data Analytics with GenAI

February 2026

What Problem Are We Solving ?

- Superstore runs yearly Gold Membership campaign
- Offers 20% discount on all purchases
- Only 14.91% customers say YES — Why so low ?

Our Goal:

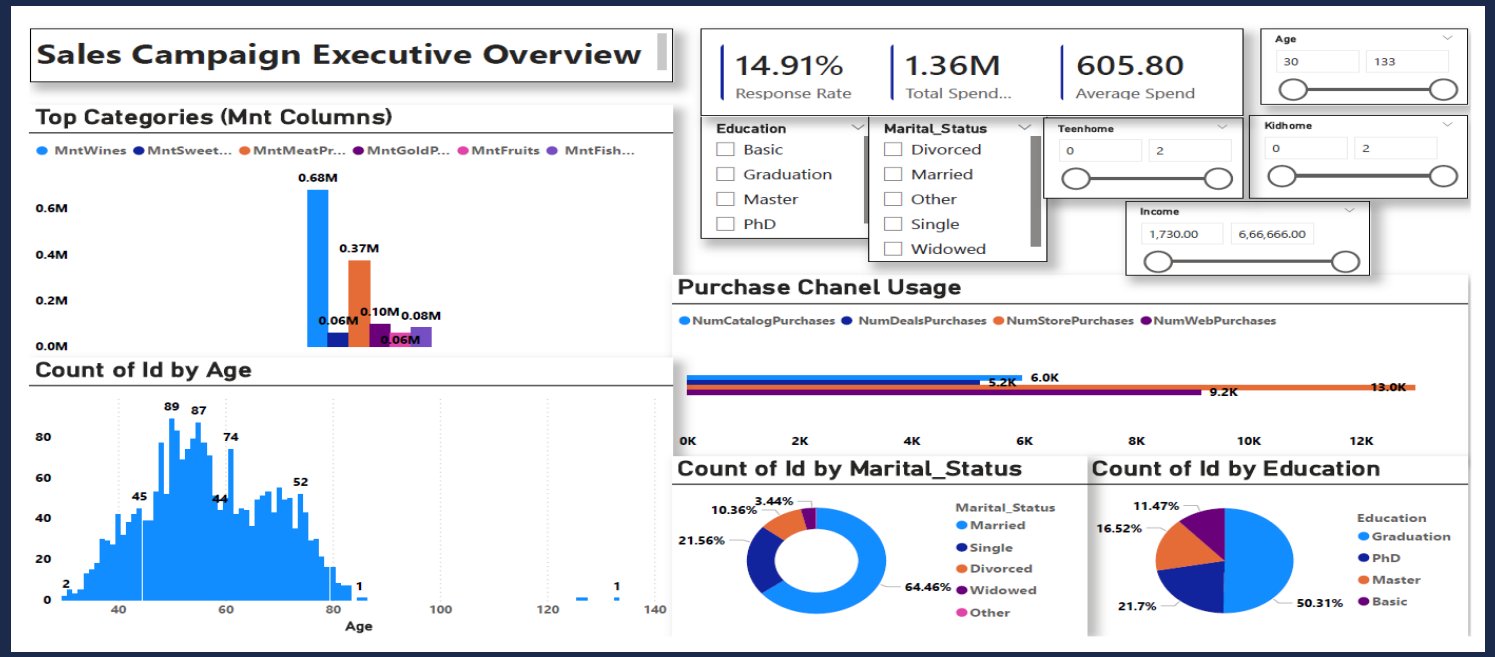
→ Find WHO will accept

→ Find WHAT drives spending

→ Improve next campaign

Campaign Performance at a Glance

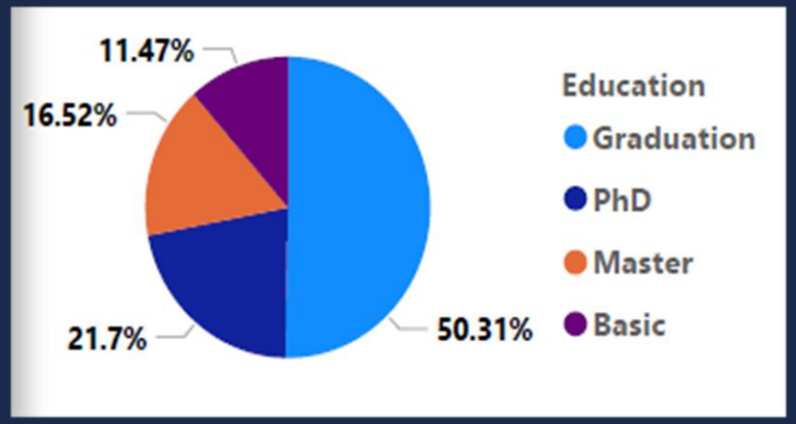
14.91%	$1.36M	$605.80	2,240
Response Rate	Total Revenue	Average Spend	Total Customers



Live Power BI Executive Dashboard — Interactive filters available

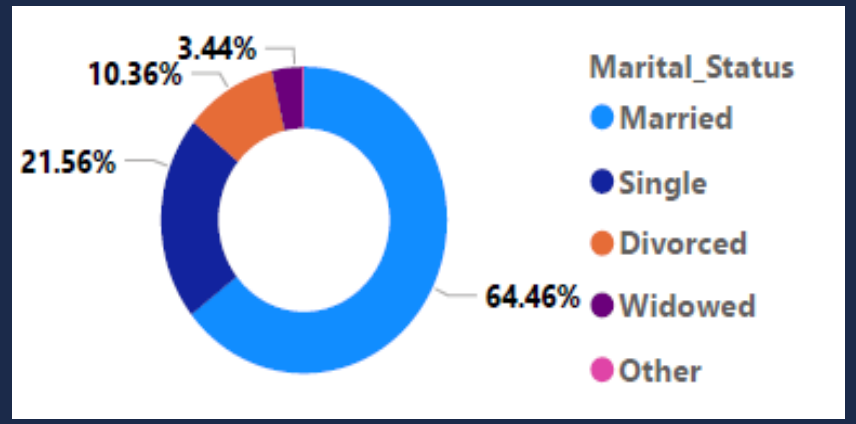
Customer Demographics — 2,240 Customers

Education Distribution



50% Graduates | 22% PhD holders

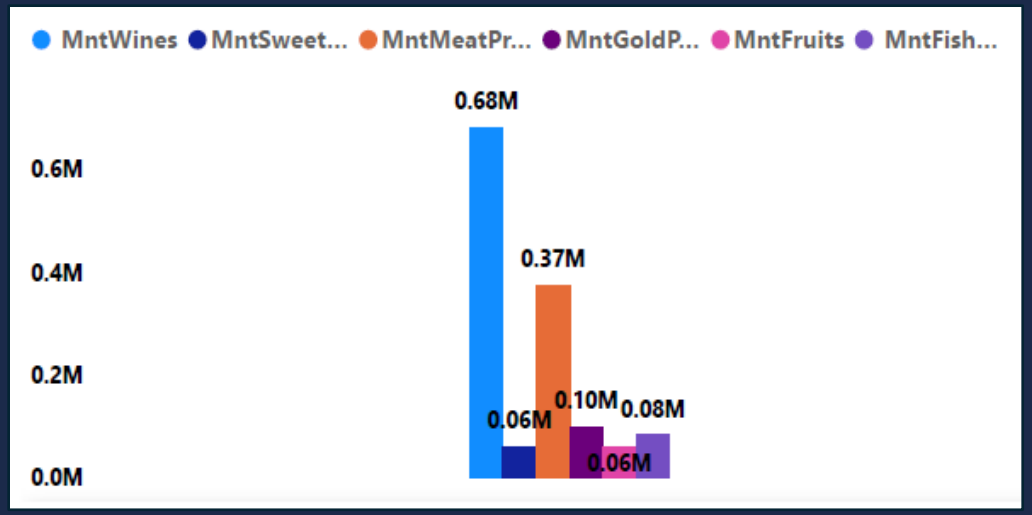
Marital Status



64% Married | 22% Single

Majority Age Group: 40-70 years

Spending Pattern — Wine & Meat Dominate



Wine	**$680,000 (50%)**
Meat	**$370,000 (27%)**
Combined	**77% of ALL revenue**

Insight: Target Wine & Meat lovers first!

Three Types of Customers

LOW SPENDER
Below $500
1,245 customers
55.6%

→

MEDIUM SPENDER
$500 — $1,500
749 customers
33.4%

→

HIGH SPENDER
Above $1,500
246 customers
11%

⭐ **Priority Target**
These 246 drive maximum ROI

Biggest Insight: Children Change Everything

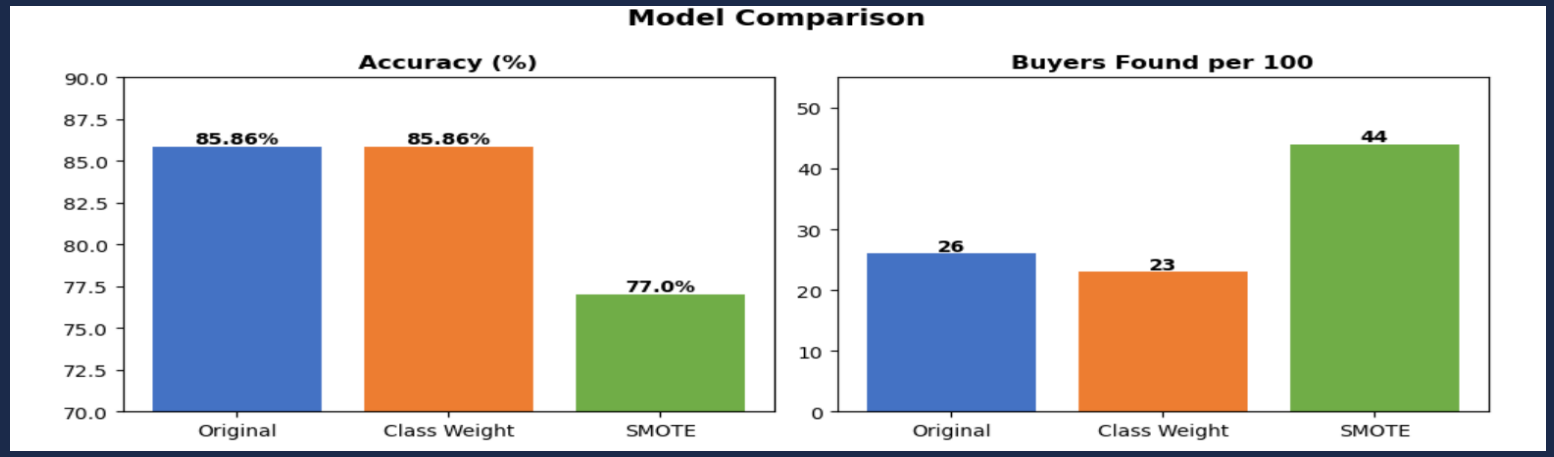
WITH CHILDREN
Avg Spend: $406
Acceptance: 10.3%

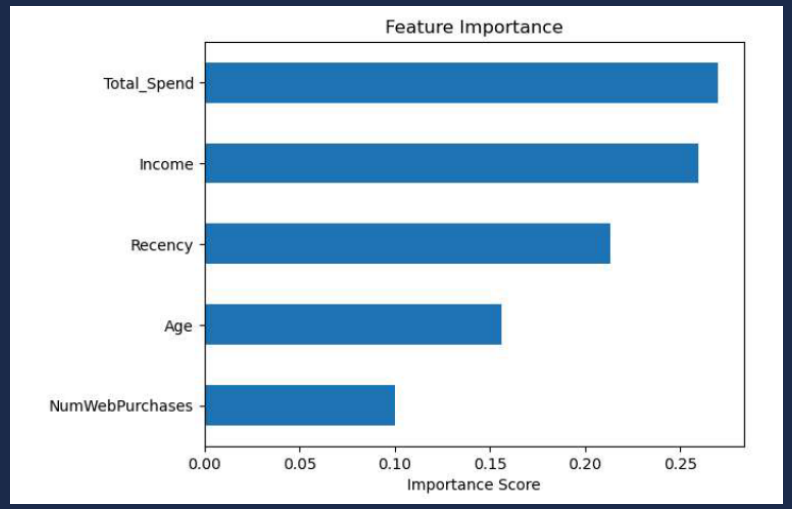
2.7x Difference
p-value < 0.001

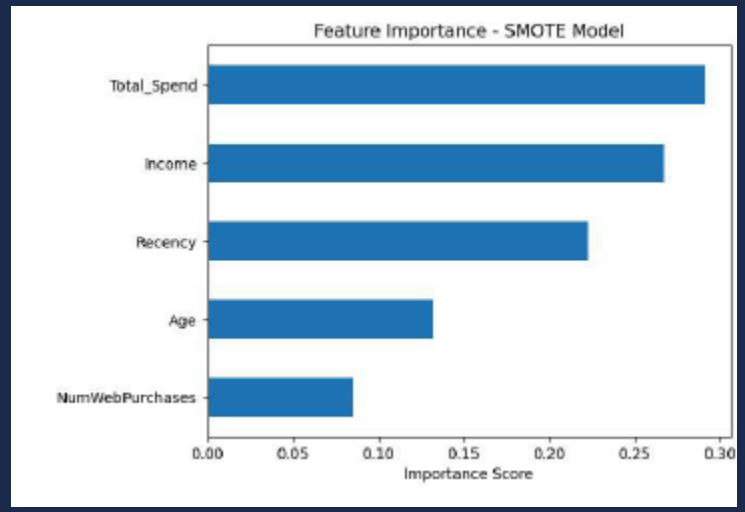
WITHOUT CHILDREN
Avg Spend: $1,106
Acceptance: 26.5%

Current membership does NOT appeal to families

Machine Learning Model — Predicting Acceptance







85.86%

Model Accuracy
Random Forest

Top Predictors:
Total Spend
Income
Recency

Model Improvement: Solving Class Imbalance

```
Model Accuracy: 85.86%

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.96      0.92       575
           1       0.52      0.26      0.34        97

    accuracy                           0.86       672
   macro avg       0.70      0.61      0.63       672
weighted avg       0.83      0.86      0.84       672
```

Base Random Forest (Accuracy): 85.86%

```
=== Before SMOTE ===
Class 0: 1331, Class 1: 237
=== After SMOTE ===
Class 0: 1331, Class 1: 1331

=== MODEL 3: SMOTE ===
Accuracy: 77.23%
              precision    recall  f1-score   support

           0       0.90      0.83      0.86       575
           1       0.30      0.44      0.36        97

    accuracy                           0.77       672
   macro avg       0.60      0.64      0.61       672
weighted avg       0.81      0.77      0.79       672
```

Optimized with SMOTE (Recall): 44%

- **Technique Used:** Applied **SMOTE** (Synthetic Minority Over-sampling Technique) to generate synthetic samples for the minority class (Buyers).
- **Strategic Shift:** Prioritized **Recall over Accuracy** to ensure fewer potential leads are missed during the campaign.
- **Business Impact:** Identifying 44% of buyers (vs 26%) allows for a more effective and high-ROI marketing outreach.

6 Actions to Boost Membership Uptake

1. Email PhD & Masters
856 customers
20.78% acceptance rate

2. Family Gold Bundle
Food category discounts
Target 10.3% gap

3. Wine & Meat Bundle
"Gourmet Food Club"
77% revenue focus

4. Win-back inactive
Recency > 60 days
Special catalog offer

5. Digital Conversion
10% first online order
1,563 in-store shoppers

6. VIP for High Spenders
Personal outreach
246 customers — top ROI

Thank You

Model optimized for **Recall (44%)** to maximize customer reach
PhD customers = Best target segment
High Spenders = Most Likely to Accept
6 Data-driven recommendations delivered

Excel | MySQL | Python | Power BI

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Career 247 — Capstone Project
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